

Chapter 4 – Product & Portfolio Management

$$\text{Year over Year (YoY) Growth} = \frac{\text{Current Value}}{\text{Prior Value}} - 1$$

$$\text{Compound Annual Growth Rate (CAGR)} = \left(\frac{\text{Final Value}}{\text{Starting Value}} \right)^{\frac{1}{N}} - 1$$

$$\text{Cannibalization Rate (\%)} = \frac{\text{Sales Lost from Existing Products (\#, \$)}}{\text{Sales of New Product (\#, \$)}}$$

Chapter 5 – Customer Profitability

$$\text{Retention Rate} = \frac{\text{Customers at End} - \text{New Customers}}{\text{Customers at Start}} \times 100$$

$$\text{Customer Lifetime Value (\$)} = \text{Margin (\$)} * \frac{\text{Retention Rate (\%)}}{1 + \text{Discount Rate (\%)} - \text{Retention Rate (\%)}}$$

$$\text{Average Acquisition Cost (\$)} = \frac{\text{Acquisition Spending (\$)}}{\text{Number of New Customers Acquired (\#)}}$$

$$\text{Average Retention Cost (\$)} = \frac{\text{Retention Spending (\$)}}{\text{Number of Customers Retained (\#)}}$$